

Sec 4.7 – L'Hopital's Rule, Growth, and Dominance

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

When we evaluated limits like $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ or $\lim_{x \rightarrow \infty} \frac{x^2 + 5}{x - 2}$ we had algebraic ways to simplify the limits so we could use limit laws and evaluate the limits in ways that made sense:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x + 3)(x - 3)}{x - 3} = \lim_{x \rightarrow 3} \frac{x + 3}{1} = \frac{3 + 3}{1} = 6$$

$$\lim_{x \rightarrow \infty} \frac{x^2 + 5}{x - 2} = \lim_{x \rightarrow \infty} \frac{x + \frac{5}{x}}{1 - \frac{2}{x}} = \frac{\lim_{x \rightarrow \infty} x + \frac{5}{x}}{\lim_{x \rightarrow \infty} 1 - \frac{2}{x}} = \infty$$

In the last case, we can see that the numerator gets bigger and bigger while the denominator goes to 1 and we can use our understanding of numbers to know that the overall fraction gets bigger and bigger, hence going to infinity.

But suppose we want to calculate the **exact** value of the limit: $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$

Substituting $x = 0$ gives us $\frac{0}{0}$, which is undefined: $\frac{e^{2(0)} - 1}{0} = \frac{1 - 1}{0} = \frac{0}{0}$

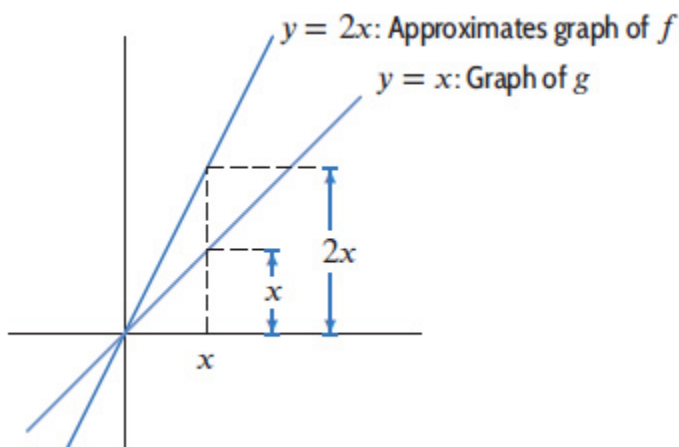
We could estimate or approximate the value of the limit by plugging in values near 0, but we can use Local Linearity, from Sec 3.9, to calculate the limit exactly.

We let $f(x)$ be the numerator, $f(x) = e^{2x} - 1$, and $g(x)$ be the denominator, $g(x) = x$.

The tangent line approximation for $f(x)$ around $x = 0$ is then $f(x) \approx f(a) + f'(a)(x - a) \rightarrow f(0) + f'(0)(x - 0)$

$f(0) = 0$, $f'(x) = 2e^{2x}$, so $f'(0) = 2$, which gives $f(x) \approx 0 + 2(x - 0) = 2x$

As we zoom in for x near 0, $\frac{f(x)}{g(x)} = \frac{e^{2x} - 1}{x} \approx \frac{2x}{x} = \frac{2}{1} = \frac{f'(0)}{g'(0)}$



So the ratio $\frac{e^{2x} - 1}{x}$ is approximated by the ratio of slopes as we zoom in.

This leads to L'Hopital's Rule.

Theorem

If f and g are differentiable, a is either a real number or $\pm\infty$, and either

$$(a) \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \quad \text{OR} \quad (b) \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \pm\infty$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Use L'Hopital's Rule to evaluate the limit $\lim_{x \rightarrow 0} \frac{\sin x}{e^x + x - 1}$

$\sin 0 = 0$ and $e^0 + 0 - 1 = 0$, so we have the form $\frac{0}{0}$, and we can use L'Hopital's Rule:

$$\lim_{x \rightarrow 0} \frac{\sin x}{e^x + x - 1} = \lim_{x \rightarrow 0} \frac{\cos x}{e^x + 1} = \frac{\cos 0}{e^0 + 1} = \frac{1}{2}$$

Partner up

Use L'Hopital's Rule to evaluate the limits, if applicable. (Remember to write LH where you are using the rule.)

1. $\lim_{x \rightarrow 0^+} \frac{\sin x}{x^2}$

2. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$

3. $\lim_{x \rightarrow \infty} \frac{\sin x}{x^2}$

4. $\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$

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$$\lim_{x \rightarrow 0} \frac{\sin x}{x^2}$$

Solution: $\lim_{x \rightarrow 0^+} \frac{\sin x}{x^2} = \lim_{x \rightarrow 0^+} \frac{\cos x}{2x}$

Here we see that $\lim_{x \rightarrow 0^+} \cos x = 1$ and $\lim_{x \rightarrow 0^+} 2x = 0$, so we are dividing

a nonzero number by a positive number (since we are coming from the right) that gets smaller and smaller, so the fraction is getting

bigger and bigger. This gives a final answer of $\lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \infty$.

(If x was approaching 0 from both sides, the limit would not exist since it would go to $-\infty$ from the left and $+\infty$ from the right.)

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin x \cos x}{2x} = \lim_{x \rightarrow 0} \frac{2 \cos^2 x - 2 \sin^2 x}{2} = \frac{2}{2} = 1$$

Here we use L'Hopital *twice* since the limit was still of form $\frac{0}{0}$ after the first application.

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x^2} = 0$$

Solution: As discussed earlier, this is not indeterminate. The numerator stays between -1 and 1 and the denominator gets bigger and bigger.

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$$

$$\textbf{Solution: } \lim_{x \rightarrow \infty} \frac{x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{2x}{e^x} = \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0,$$

where the final answer follows since we are dividing a finite positive number by a value that gets bigger and bigger, giving a smaller and smaller positive fraction.

The last example showed us that as $x \rightarrow \infty$, e^x grows much faster than x^2 . We say that e^x **dominates** x^2 .

Definition

For positive functions f and g , we say g dominates f as $x \rightarrow \infty$ if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0.$$

Which function dominates the other: $\ln x$ or x ?

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0, \text{ so } x \text{ dominates } \ln x.$$